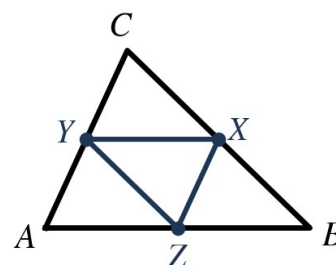
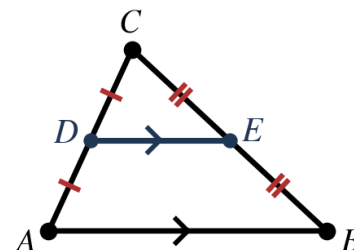


Speak Like A Mathematician:

midsegment of a triangle:

midsegment triangle:



The Triangle Midsegment Theorem

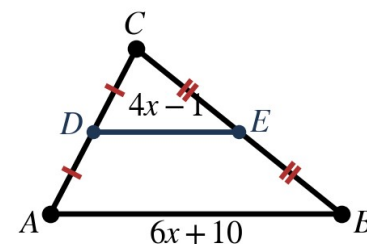
The segment connecting the midpoints of two sides of a triangle is _____ to the third side and is _____ as long as that side.

Every triangle has _____ midsegments. Together they form the _____.

Example 1 - Use the Midsegment Theorem

In $\triangle ABC$, D is the midpoint of $\overline{AC}$ and E is the midpoint of $\overline{BC}$.

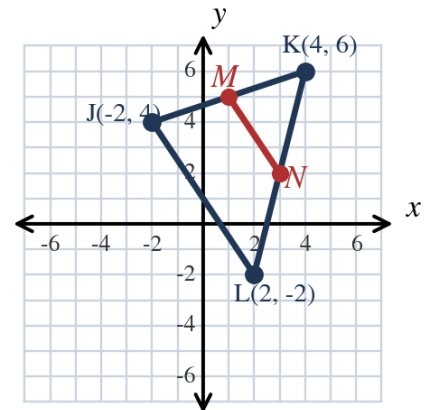
- Write an equation relating DE and AC.
- Solve for x.
- Find DE and AC, then check.



Example 2 - A Midsegment in the Coordinate Plane

In $\triangle JKL$ with $J(-2, 4)$, $K(4, 6)$, and $L(2, -2)$, M is the midpoint of $\overline{JK}$ and N is the midpoint of $\overline{KL}$. Show that $\overline{MN} \parallel \overline{JL}$ and that $MN = \frac{1}{2} JL$.

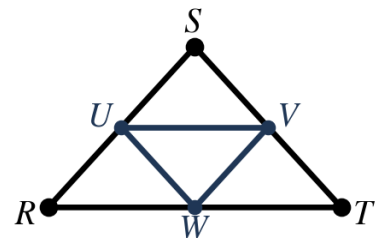
- Find the coordinates of M and N .
- Find the slope of $\overline{MN}$ and the slope of $\overline{JL}$.
- Find MN and JL .



Example 3 - A Roof Truss

In the truss, U , V , and W are the midpoints of the sides of $\triangle RST$. $RT = 90$ in. and $VW = 57$ in.

- Find UV .
- Find RS .
- How does the perimeter of $\triangle UVW$ compare with the perimeter of $\triangle RST$?



Example 4 - A Jogging Route

Maple Street connects the midpoints of Oak Street and Birch Street. Oak is 1.4 mi, Birch is 1.8 mi, and Cedar — the third side — is 2.2 mi. You start at your home at the midpoint of Oak, jog along Maple to the midpoint of Birch, down Birch to the corner, along Cedar, and back up Oak to home.

- a) How long is Maple Street?
- b) How far do you jog on Birch and on Oak?
- c) Find the total distance.

Example 5 - Reasoning with Midsegments

Answer each question about the midsegment triangle.

- a) $\triangle ABC$ has sides of 10, 14, and 18. Find the perimeter of its midsegment triangle.
- b) A midsegment measures 12. What is the length of the side it is parallel to?
- c) Why is the midsegment triangle always similar to the original triangle?
- d) Could a midsegment ever be longer than the side it is parallel to?